

Subject Name: Metabolic Engineering**Subject No:** BT41014**Prerequisites:** Microbiology (BT20205) and Biochemistry (BT20203)

Objective: Metabolic engineering is an emerging field of biotechnology/bioprocess engineering which aims towards purposeful modification of cellular (metabolic, gene regulatory, and signaling) processes/networks to achieve desirable goals such as enhanced production of metabolites including pharmaceuticals, biofuels and biochemicals and other biotechnology products. This course aims to provide fundamental and advanced knowledge in the development of microbial strain for bioproduction through metabolic engineering.

L-T-P 3-0-0**Credit 3****Content:**

1. Introduction to Metabolic Engineering, Basic concepts; Scopes and Applications - 2 Lectures
2. Metabolism overview. Cellular Transport processes, Fueling Reactions - 3 Lectures
3. Biosynthetic reactions, Polymerization, Growth Energetics); Regulation of Metabolic Pathways - 4 Lectures
4. Reconstruction of Genome-scale metabolic network - 4 lectures
5. Examples of pathway manipulations by metabolic engineering: Ethanol, Aminoacids, antibiotics, vitamins, biopolymers, etc.- 4 Lectures
6. Examples of pathway manipulations by metabolic engineering: Improvements of cellular properties, Biodegradation - 4 Lectures
7. Metabolic modeling; Material balance and data consistency - 3 Lectures
8. Metabolic Flux Analysis: Flux Balance Analysis (FBA), Flux Variability and Control Analysis, Flux Map - 6 Lectures
9. Experimental determination of Metabolic Fluxes: Isotope labeled substrate, Isotope mapping Mapping Matrix, Isotope Distribution Vector - 6 Lectures
10. Application of metabolic Flux Analysis - 2 Lectures

Text Books:

1. Metabolic Engineering, Principles and Methodologies; G N Stephanopoulos, A A Aristidou, J Nielsen (1998).

Reference Books:

1. Advances in Biochemical Engineering/Biotechnology; Metabolic Engineering, Volume Editor: Jens Nielsen
2. Systems Metabolic Engineering, Methods and Protocols; H S Alper
3. Metabolic Pathway design, A Practical Guide; P Carbonell